

Saaketh Koundinya Gundavarapu

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 Portfolio |  GitHub |  Google Scholar

EDUCATION

New York University

Master of Science in Computer Engineering, GPA: 3.97/4.0 | Dean's Scholarship: \$18,000

New York, NY

Aug 2022 – May 2024

National Institute of Technology, Warangal

Bachelor of Technology CGPA: 8.38/10

Warangal, India

Aug 2015 – May 2019

TECHNICAL SKILLS

Quantitative: Stochastic Calculus, Monte Carlo Simulation, Probability Theory, Regression Analysis, Convex Optimization, Linear Programming/QP/MILP, Rockafellar-Uryasev, Dynamic Programming, Diffusion Models, Machine Learning, Deep Learning.

Programming: Python (primary), MATLAB, C++, TypeScript, React (Next.js), SQL.

Infrastructure & Data: PyTorch, AWS (EC2, S3, Lambda, Amplify, Elastic Beanstalk), MongoDB, MySQL.

Power Market: BESS Co-Optimization, OTC/Bespoke Trade & Hedge Valuation, NG Transport Intrinsic/Extrinsic Decomposition, Portfolio Optimization, Delta Hedging, Power Derivatives, Risk Management, Market Making, Markets (ERCOT, CAISO, PJM), Retail Valuation.

EXPERIENCE

Quantitative Analyst & Developer

Aug 2024 – Present

Scoville Risk Partners – North American Power, Natural Gas & BESS Markets

Princeton, NJ

- **BESS Pricing & Daily Mark-to-Market:** Primary quant responsible for daily MtM valuation of **2GW+** of BESS assets across ERCOT and CAISO. Built and productionized a co-optimized dispatch engine across Day-Ahead, Real-Time, and Ancillary Services, valuation of 20-year battery portfolios in minutes for risk management and origination workflows.
 - * **BESS Trading Strategy:** Developed a Day-Ahead power price forecasting model and systematic trading strategies that captured **90–110%** of daily spreads in backtesting.
- **Hedge Lab – Power Portfolio Hedging:** Built and deployed Hedge Lab in production across multiple retail and utility clients; optimized power portfolios against listed market instruments and OTC hedges achieving **\$100M+ reduction in volatility and improvement in VaR.**; evaluated structured deals across \$10M+ in notional trade flow, providing quantitative what-if analysis and quotes to support client deal execution.
- **Natural Gas Storage & Transport:** Primary quant for the real-time Natural Gas Transport Valuation Platform, responsible for valuation modeling and dashboard development; also built Storage dashboards. **Marked 100+ transport assets to market daily**; modeled storage intrinsic/extrinsic value decomposition, P&L explain, and documentation.
- **Retail Energy Book Valuation:** Engineered a scalable valuation platform pricing and risk-managing **1M+ electricity and gas meters monthly** across the U.S.; modeled each account individually & performed valuation analysis and P&L attribution to support trading desk decisions.
- **Systematic Market-Making & Risk Platform (POC Stage)** formulated a systematic market-making and risk framework for dealers, retailers, and BESS participants across ERCOT, and CAISO; developed frameworks for continuous hedging of the book.

Graduate Research Assistant

Jan 2023 – May 2024

New York University – DICE Lab & TA - Deep Learning (Yann LeCun)

New York, NY

- **Machine Unlearning in LLMs** (arXiv:2405.15152, 10 citations): Developed methods to selectively remove knowledge from fine-tuned LLMs without full retraining via gradient-based unlearning and activation steering.
- **Zero-Shot Object Segmentation (SLIP)** (arXiv:2405.07284, 8 citations): Combined Segment Anything Model (SAM) with CLIP for zero-shot segmentation on 10k COCO subset — improved accuracy from 34% to 69%.
- **Teaching:** Grader and TA for graduate courses: Deep Learning (LeCun & Canziani) and Inference & Representation.

Software Development Engineer

July 2019 – July 2022

SAP Labs India – Full-Stack, Enterprise Healthcare & Supply Chain

Bangalore, India

- Led end-to-end development of a production supply chain application, in production at **25 pharmaceutical companies**; received Emerging Talent award (Demo).
- Designed reusable UI component library; improved application load time by **80%** across 4 legacy systems; built real-time business document tracking infrastructure. (JavaScript, ABAP, HANA, UI5)

PROJECTS

Vision-KAN: Vision Transformers with KAN Layers | PyTorch, KAN, Open-source

May 2024

- Replaced MLP layers with Kolmogorov-Arnold Network (KAN) layers in Vision Transformers – **1,000+ 📄 downloads and 250 ★ on GitHub within 2 weeks**; achieved **86.14% on ImageNet** and **96.16% on CIFAR-100**.

LLMs for Personalised Recommendation | Pytorch, PEFT, LoRA, DPO, Llama

Jan 2024 – May 2024

- Implemented conditional recursive classification with LLMs – **reduced the recommendation space from 10,000 to 100 items**.